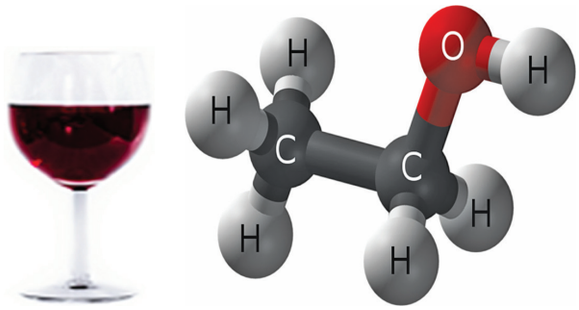


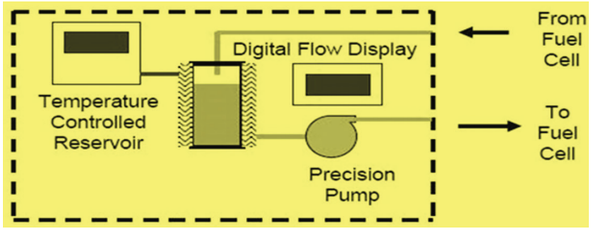
Figure 3.2.1.1 Alcohol Chemical Bond

3.2.1.2 Methanol

Methanol, also known as wood alcohol, can be used as an alternative fuel. M85 (a blend of 85 % methanol and 15 % gasoline) and M10 (a blend of 10% methanol and 90% gasoline) are used as fuel and emissions are lower than conventional vehicles. It has high octane number. Methanol is cheap to produce and has a lower risk of flammability when compared to petrol. The cost of fuel is low. However, methanol is corrosive. Fig 3.2.1.2, 3.2.1.2(a) shows Storage of Methanol and Methanol Unit respectively.



Figure 3.2.1.2 Storage of Methanol



Methanol-unit

Figure 3.2.1.2(a) Methanol Unit

3.2.1.3 Ethanol

Ethanol is also called as Ethyl alcohol. Ethanol can be produced by fermenting and distilling crops such as corn, barley or wheat. In India, Ethanol is extracted from molasses of sugarcane. It can be blended with gasoline to increase octane levels and improve emissions quality. E85 (a blend of 85 % ethanol and 15 % gasoline) and E10 (a blend of 10% ethanol and 90% gasoline) are used as fuel. Fig 3.2.1.3 shows the Carbon Cycle.

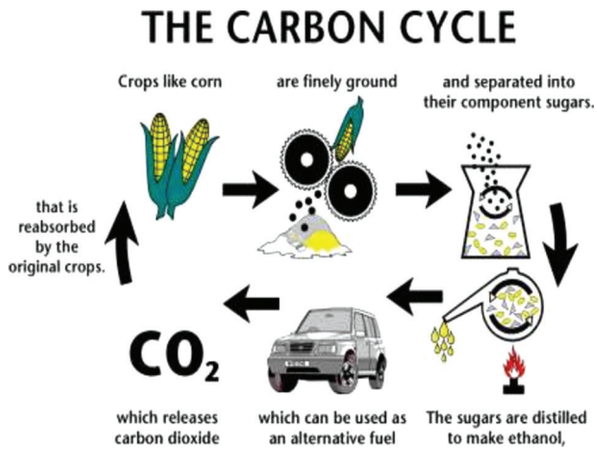


Figure 3.2.1.3 The Carbon Cycle

3.2.1.4 Bio Diesel

Bio diesel is a domestically produced, renewable fuel that can be manufactured from vegetable oils, animal fats, for use in diesel vehicles. Biodiesel can also be blended with diesel and used in unmodified engines. B20 (i.e 20% biodiesel blended with diesel) is a most common biodiesel blend. B20 has good balance of cost, emissions, cold-weather performance, materials compatibility, and ability to act as a solvent. Biodiesel is safe, biodegradable, reduces air pollutants associated with vehicle emissions, such as particulate matter, carbon monoxide and hydrocarbons. Fig 3.2.1.4 shown Bio Diesel Production Process Cycle.

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